Code Screenshots

```
static class Product extends Item implements Stock
    {
        private int quantity;
        private double price;

        Product(int productId, String productName, int quantity, double price)
        {
            super(productId, productName);
            this.quantity = quantity;
            this.price = price;
        }

        public int getProductId()
        {
            return productId;
        }

        public String getProductName()
        {
            return productName;
        }

        public int getQuantity()
        {
            return quantity;
        }

        public double getPrice()
        {
            return price;
        }

        public void setProductName(String productName)
        {
            this.productName = productName;
        }

        public void setQuantity(int quantity)
        {
            this.quantity = quantity;
        }

        public void setPrice(double price)
        {
            this.price = price;
        }
```

```
        public String getStockStatus()
        {
            if (quantity == 0)
                return "Out of Stock";
            else if (quantity <= 5)
                return "Low Stock";
            else
                return "Available";
        }
```

```
static void addProduct()
    {
        int id = readInt("Enter Product ID: ");

        if (id <= 0)
        {
            System.out.println("Product ID must be positive.");
            return;
        }

        if (productIds.contains(id))
        {
            System.out.println("Product ID already exists!");
            return;
        }

        String name = readString("Enter Product Name: ");
        int quantity = readInt("Enter Quantity: ");
        double price = readDouble("Enter Price: ");

        if (quantity < 0 || price < 0)
        {
            System.out.println("Quantity and price cannot be negative.");
            return;
        }

        Product p = new Product(id, name, quantity, price);

        products.add(p);
        productIds.add(id);
        productMap.put(id, p);

        saveData();
```

```
        System.out.println("Product added successfully!");
    }
```

```
static void saveData()
    {
        try (PrintWriter pw = new PrintWriter(new FileWriter(FILE_NAME)))
        {
            for (Product p : products)
            {
                pw.println(
                    p.getProductId() + "|" +
                    p.getProductName() + "|" +
                    p.getQuantity() + "|" +
                    p.getPrice()
                );
            }
        }
        catch (IOException e)
        {
            System.out.println("Error saving inventory: " + e.getMessage());
        }
    }
```

Output Screenshots

```
========== INVENTORY MANAGEMENT SYSTEM ==========
1. Add Record
2. View All Records
3. Search Record
4. Update Record
5. Delete Record
6. Stock Report
7. Exit
Enter your choice:
```

```
========== INVENTORY MANAGEMENT SYSTEM ==========
1. Add Record
2. View All Records
3. Search Record
4. Update Record
5. Delete Record
6. Stock Report
7. Exit
Enter your choice: 1
Enter Product ID: 101
Enter Product Name: washing machine
Enter Quantity: 10
Enter Price: 200000
Product added successfully!
```

```
========== INVENTORY MANAGEMENT SYSTEM ==========
1. Add Record
2. View All Records
3. Search Record
4. Update Record
5. Delete Record
6. Stock Report
7. Exit
Enter your choice: 3
Enter Product ID to search: 101
-----------------------------------------
Product ID   : 101
Product Name : washing machine
Quantity     : 10
Price        : 200000.0
Stock Status : Available
-----------------------------------------
```

```
========== INVENTORY MANAGEMENT SYSTEM ==========
1. Add Record
2. View All Records
3. Search Record
4. Update Record
5. Delete Record
6. Stock Report
7. Exit
Enter your choice: 4
Enter Product ID to update: 101
1. Update Product Name
2. Update Quantity
3. Update Price
4. Update All Details
Enter your choice: 2
Enter New Quantity: 8
Product updated successfully!
```

```
========== INVENTORY MANAGEMENT SYSTEM ==========
1. Add Record
2. View All Records
3. Search Record
4. Update Record
5. Delete Record
6. Stock Report
7. Exit
Enter your choice: 6

Generating Stock Report...

========== STOCK REPORT ==========
Total Products : 1
Available      : 1
Low Stock      : 0
Out of Stock   : 0
==================================
```